

Experiments and Randomized Controlled Trials

Random assignment, treatment/control groups, and causal claims

WEEK 8 • DAY 1

Two questions for today

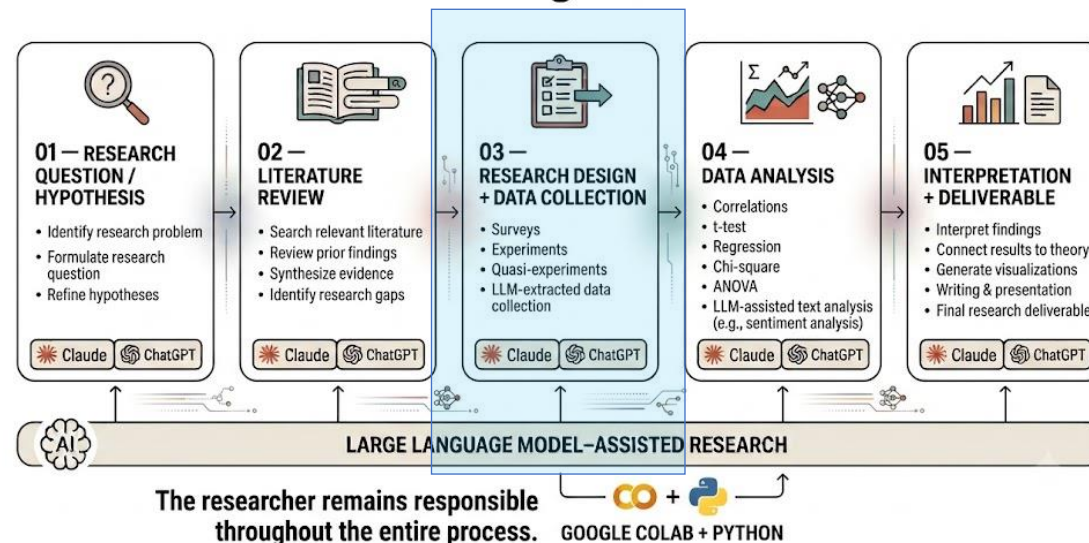
- How is an experiment different from a survey study?
- What does random assignment do in an experiment?
- What can an RCT help us say about cause and effect?

RCT = randomized controlled trial: a study that compares conditions after random assignment.

Connect to what we already learned

- Week 6: RQs, hypotheses, variables, measurement, and survey as a research design
- Week 7: sampling and measurement quality.
- Now: research design—how the study is organized to answer the question.
- Experiments are one design that can test causal hypotheses.

Research Methods in the Age of AI



Association is not the same as causation

Association / correlation (Survey)

- Two variables move together.
- Example: people who use TikTok more also report more news exposure.
- This does NOT prove TikTok **caused** the news exposure.

Causation (Experiments: RCT only)

- One factor produces a change in another factor.
- Example: seeing an AI label causes lower credibility ratings.
- **Requires stronger design evidence.**

Experiments are designed to strengthen causal evidence.

	Survey	RCT / Experiment
Main purpose	Describe attitudes, behaviors, or relationships	Test whether something <u>causes</u> a change
What researcher does	<u>Measures</u> variables as they already exist	<u>Manipulates</u> something
Random assignment?	Usually no	Yes — participants are randomly assigned to conditions
Example	Ask people how often they see influencer ads and how much they trust influencers	Randomly show one group an ad with #Ad and another group the same ad without #Ad
What can we conclude?	“These variables are related”	Stronger evidence that the treatment caused the outcome

The causal question behind an experiment

- Would the same person have responded differently under a different condition?
- We cannot show one person both versions at the same exact moment.
- So, we compare groups that are made as similar as possible.
- Random assignment helps create comparable groups.

Causal logic: What would have happened otherwise?

What is an experiment?

- The researcher controls or manipulates something.
- Participants are assigned to different conditions.
- The researcher measures an outcome after exposure to the condition.
- The goal is to test whether the condition affects the outcome.

Experiment = manipulate a cause, then measure an outcome.

What is an RCT?

- RCT = Randomized Controlled Trial.
- Participants are randomly assigned to at least two conditions.
- One condition usually receives the treatment or intervention.
- Another condition serves as a comparison or control.
- Then outcomes are compared across groups.

Random assignment is the key feature.

Simple communication example

Treatment condition

- Participants see an influencer post with an #Ad disclosure.
- This is the version we think may affect credibility.

Control condition

- Participants see the same post without the #Ad disclosure.
- Everything else should stay as similar as possible.

Outcome: perceived credibility measured after viewing the post.



A



B



Independent variable and dependent variable

- Independent variable: the factor the researcher changes or compares.
- Dependent variable: the outcome the researcher measures.
- Example IV: #Ad disclosure present vs. absent.
- Example DV: perceived credibility rating.

IV = possible cause. DV = possible effect.



#	Research question / hypothesis	IV / Predictor	DV / Outcome
1	Does seeing a #Ad disclosure affect perceived influencer credibility?		
2	People who see an AI-generated label will rate a news post as less trustworthy.		
3	Is frequency of TikTok news use related to perceived news credibility?		
4	Do viewers rate a video apology as more sincere than a written apology?		
5	Students who follow more sports journalists on social media will report greater knowledge of team news.		

In experiments, the IV is usually **manipulated**.
 In surveys, the IV/predictor is usually **measured**.



RCT vocabulary

Term	Meaning	Example
Treatment	The condition that includes the intervention or message feature	Post includes #Ad disclosure
Control	The comparison condition	Same post without #Ad disclosure
Outcome	What is measured after the condition	Credibility rating
Random assignment	Randomly place participants into conditions	Computer assigns each participant to one version

Know these words before you write your design section.

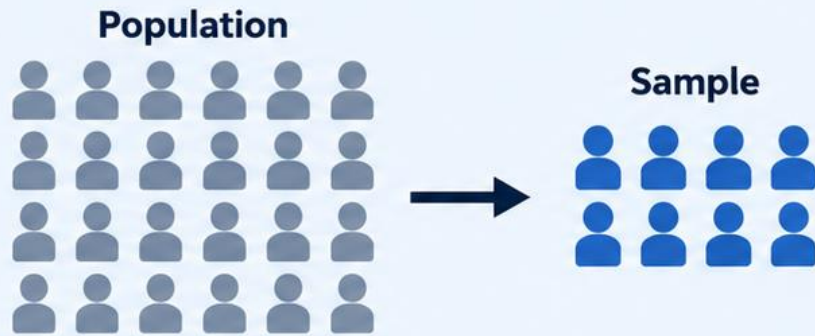
Random assignment

- Each participant has a chance of being placed in any condition.
- A computer, survey platform, or random number process can assign conditions.
- The researcher should NOT decide who gets which condition.
- The goal is to balance both known and unknown differences across groups.

Random assignment helps reduce selection bias between conditions.

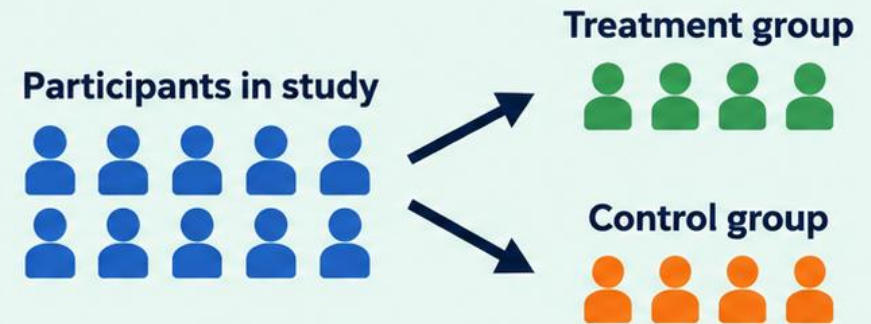
Random Sampling vs. Random Assignment

Random Sampling



- Randomly choosing people from a population to enter the study.
- **Goal:** improve representativeness and support generalization.

Random Assignment



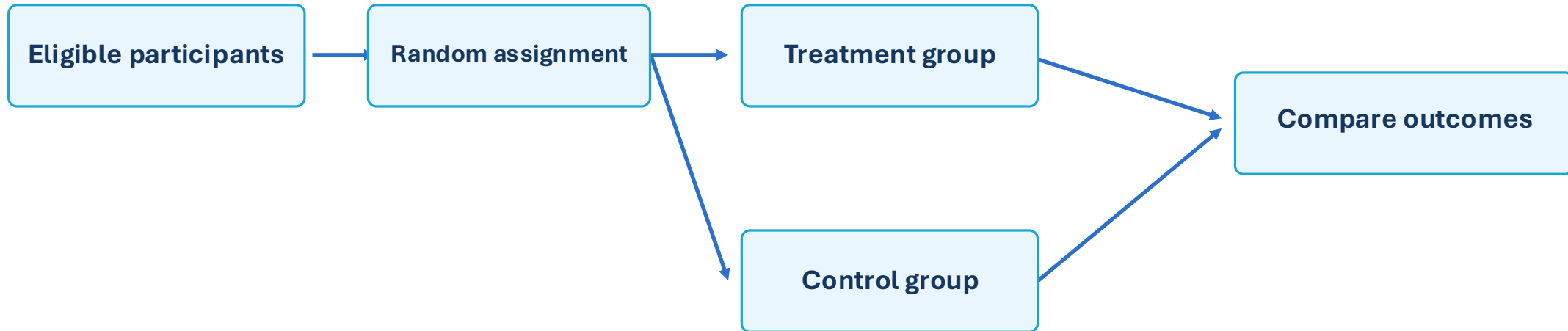
- Randomly placing participants into study conditions.
- **Goal:** make groups similar for stronger causal inference.

Key Difference

Random sampling = WHO gets into the study

Random assignment = WHICH condition they get

The basic RCT logic



The difference between groups is treated as the estimated effect of the treatment.

Why randomization matters

- Imagine the treatment group already likes the brand more before the study.
- If they rate the brand higher afterward, the effect may not be from the treatment.
- Random assignment makes it less likely that one group starts out systematically different.
- It does not guarantee perfect groups, especially with small samples, but it helps.

Randomization protects internal validity.

Holding other things constant

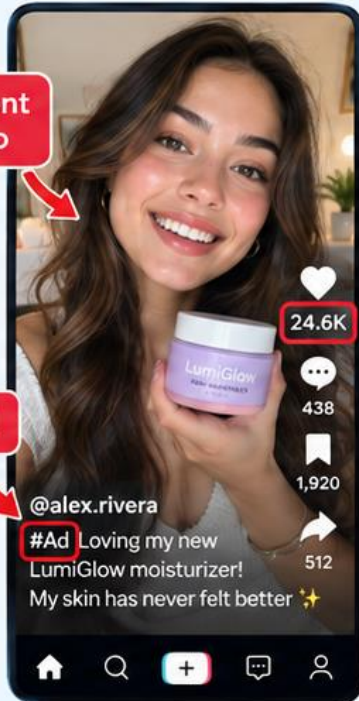
- In a clean experiment, the treatment and control versions should differ only in the key variable.
- Same influencer, same image, same caption, same platform layout.
- Only the #Ad label changes.
- If many things change at once, we cannot know which feature caused the outcome.

Change one main thing at a time when possible.

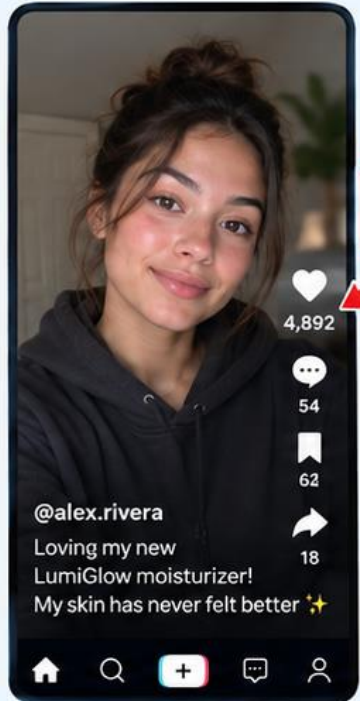
TikTok example: bad vs. better design

Bad design

Treatment



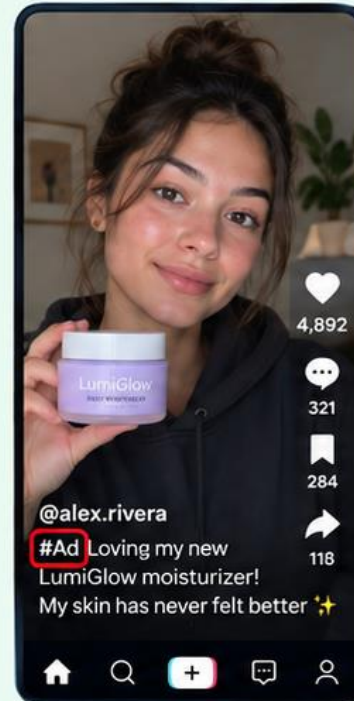
Control



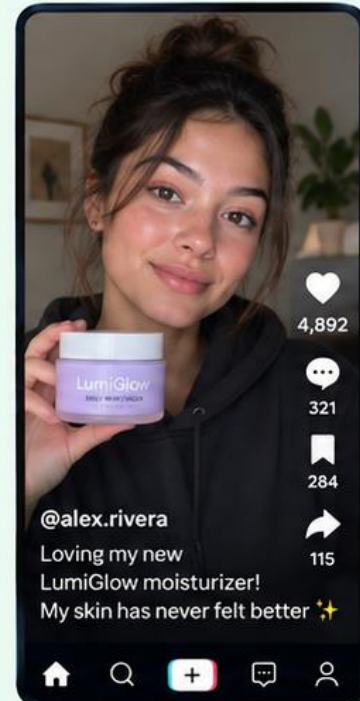
Too many things differ, so we cannot tell what caused the effect.

Better design

Treatment



Control



Now the comparison isolates the effect of the #Ad label.

A confound is an alternative explanation created by the design.

Confound: when another difference sneaks in

Bad design

- Treatment post has #Ad label.
- Treatment post also uses a more attractive photo.
- Treatment post has more likes.

Better design

- Treatment and control use the same photo.
- Same likes, same caption, same layout.
- Only #Ad label differs.

A confound is an alternative explanation created by the design.

Manipulation: what exactly changes?

- The manipulation is the feature the researcher changes across conditions.
- Example: AI-generated label vs. no label.
- Example: emotional apology vs. informational apology.
- Example: human journalist byline vs. AI-assisted byline.
- The manipulation should be visible and understandable to participants.

A weak manipulation may produce unclear results.

Manipulation check

- A manipulation check asks whether participants noticed or understood the manipulation.
- Example: “Did the post include an #Ad disclosure?” Yes / No.
- Example: “Who was described as writing the article?” Human / AI / Not sure.
- It helps you know whether the experimental condition **worked as intended**.

Manipulation checks do not replace the main outcome measure.

Outcome measurement still matters

- Even in an RCT, poor measurement can produce weak evidence.
- Use clear, neutral questions.
- Use established measures when possible ($\alpha > .7$).
- Report how the outcome was measured.
- Ask whether the measure fits your population and context.

Strong design + weak measure = still a problem.

Posttest-only vs. Pretest-posttest

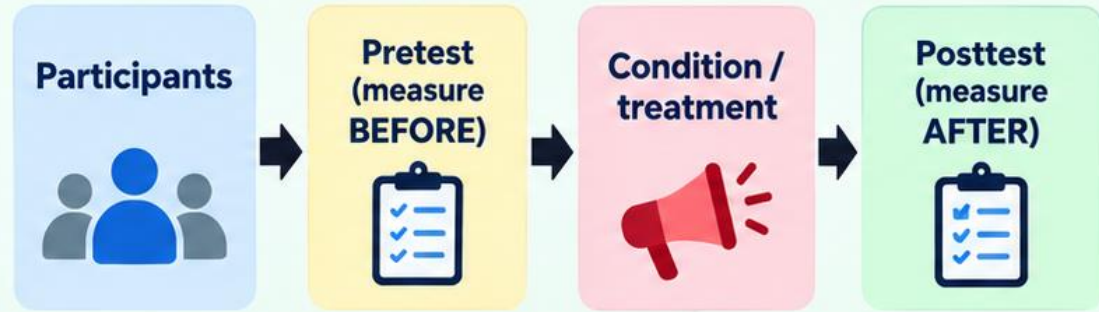
Posttest-only design



- Measure the outcome only after the condition.
- Simple design; no baseline measure.

Example: See #Ad post → then rate credibility.

Pretest-posttest design



- Measure before and after the condition.
- Helps show change over time.
- But the pretest may alert participants to what the study is about.



Possible issue:
pretest sensitization

The pretest may influence later responses.

Example: Rate credibility → see #Ad post → rate credibility again.



Key idea

Pretests can be useful because they show change from before to after.
But they can also make participants more aware of the study purpose.

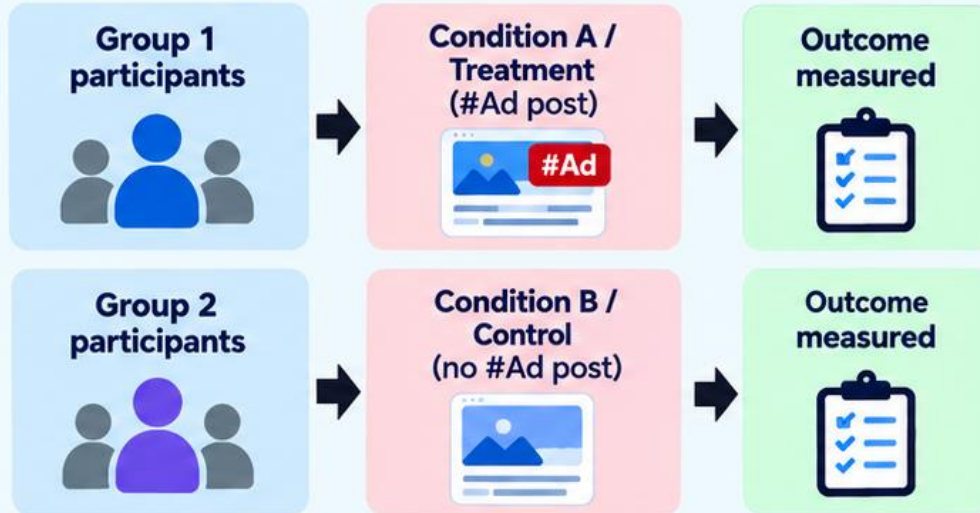
Pretest and posttest designs

- Posttest-only: measure the outcome after the condition.
- Pretest-posttest: measure before and after the condition.
- Pretests can help show change over time.
- But pretests can also alert participants to what the study is about.

Choose based on what your RQ requires.

Between-subjects vs. Within-subjects

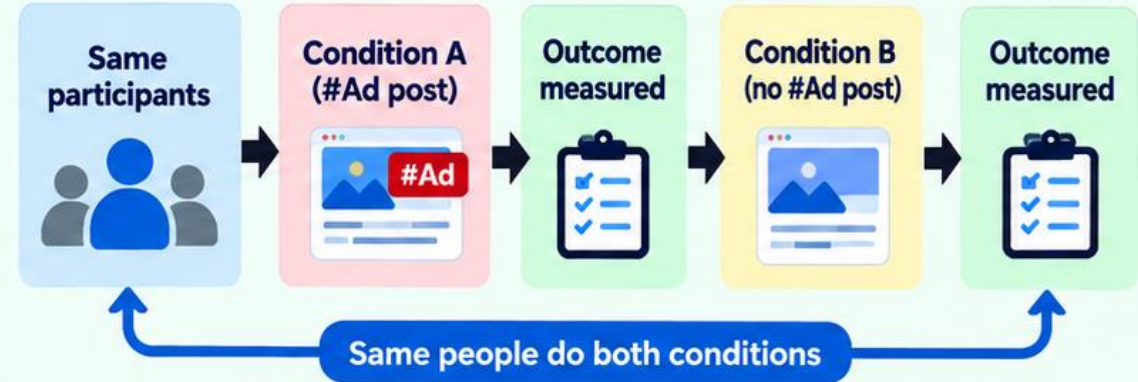
Between-subjects design



- Different participants are placed in different conditions.
- We compare one group to another group.
- Helps avoid carryover from seeing both conditions.

Example: Group 1 sees a post with #Ad; Group 2 sees the same post without #Ad.

Within-subjects design



- The same participants experience all conditions.
- Useful because people serve as their own comparison.
- Often needs fewer participants.

Possible issue:
order / carryover effects

Seeing one condition first may influence responses to the next one.

Example: The same student rates a post with #Ad and a version without #Ad.



Key idea

Between-subjects = compare different groups. Within-subjects = compare the same people across conditions.

Between-subjects vs. within-subjects

Between-subjects experiment

- Each participant sees only one condition.
- Group A sees treatment; Group B sees control.
- Common and simple for communication experiments.

Within-subjects experiment

- Each participant sees multiple conditions.
- The same person rates several message versions.
- Useful, but participants may compare or guess the purpose.

For communication research, between-subjects designs are usually easiest to explain.

A/B testing is a real-world experiment

- A/B tests compare two or more versions of a message, webpage, subject line, or design.
- Users are randomly shown version A or version B.
- The organization compares outcomes such as clicks, sign-ups, viewing time, or donations.
- A/B tests are common in advertising, journalism, UX, and platform design.

A/B test = experiment in a real-world system.

Lab experiment vs. field experiment

Type	What it looks like	Trade-off
Lab / survey experiment	Participants view controlled stimuli in a study setting	More control, sometimes less realism
Field experiment	Experiment occurs in a real-world setting such as a website or campaign	More realism, often less control

Both can be useful; the design should fit the research question.

Internal validity threats in experiments

- Confounds: conditions differ in more than the intended way.
- Demand characteristics: participants guess the purpose and change behavior.
- Attrition: people drop out differently across conditions.
- Noncompliance: participants do not receive or notice the assigned condition.

RCTs are strong, not magical.

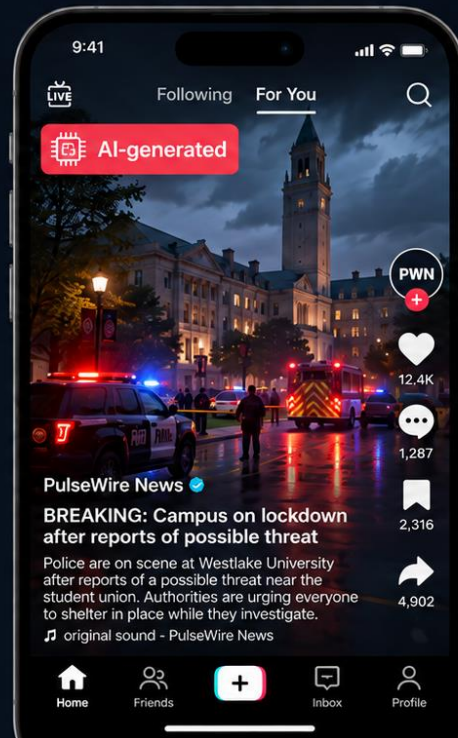
Ethics of experiments

- Do not expose participants to unnecessary harm.
- Be careful with deception; justify it and debrief when required.
- Protect privacy and sensitive data.
- Make participation voluntary.
- Ask whether the study is appropriate for the population.

Research design is also an ethical decision.

Activity: design a tiny RCT

Does labeling a breaking-news TikTok post as AI-generated reduce its credibility and willingness to share?



1. Write one hypothesis.
2. Identify the independent variable and dependent variable.
3. Describe the treatment and control conditions.
4. Write one manipulation-check question.
5. Write one outcome-measure question.

Common beginner mistakes

- Calling a survey an experiment when nothing is manipulated.
- Changing too many things between treatment and control.
- Forgetting to explain random assignment.
- Using causal language when the design only shows association.
- Choosing measures that do not match the variables.

Design wording matters.

What can I conclude?

Design	Safer conclusion	Avoid saying
Cross-sectional survey	X is related to Y	X caused Y
RCT / experiment	The manipulation affected the outcome in this study	This proves the effect for everyone
A/B test	Version A produced more clicks than Version B in this setting	Version A is always better

Match your conclusion to your design.

Assignment 4

Research Design Brief

Pair assignment • Due October 27 • 20% of course grade

Assignment 4: what you submit

- One document per pair through Blackboard.
- Approximately 4–5 double-spaced pages, excluding references.
- Use a 12-point readable font.
- Use any standard scholarly citation style, including APA, as long as it is consistent.
- Purpose: develop a concise research proposal.

Both partners normally receive the same assignment grade.

Your proposal must include: Part 1

- Research question or hypothesis: what does the study seek to understand or test?
- Research rationale and brief literature review: why does the question matter?
- Engage with at least six relevant peer-reviewed studies.
- You may use of AI for systematic review: idescribe tool, task, verification, and limitations.

If AI is not used, briefly explain why it is unnecessary for your proposed study.

Your proposal must include: Part 2

- Population and sample: who is the population, what sample will you recruit, and how?
- Key variables or measures: what will be measured and how?
- Research design: survey, experiment, A/B test, quasi-experiment, or another design.
- Strengths and limitations: identify major advantages and threats to validity.

The design should clearly fit the RQ or hypothesis.

Assignment 4 evaluation

Area	Weight
Research question/hypothesis and rationale	15%
Literature review and connection to prior research	20%
Sampling and measurement	20%
Appropriateness and justification of research design	25%
Appropriate justification and critical consideration of AI use or non-use	10%
Limitations and overall clarity	10%

The research design section carries the most weight.

Collaboration expectations

- You are expected to contribute meaningfully to pair work.
- If serious collaboration problems arise, contact me early.
- Do not wait until the assignment is due.
- I may adjust collaborative arrangements or individual grades when warranted by documented contributions.

Start early enough to identify problems before October 13.